

Supervised Learning, COMP0078
Sample Examination Paper 2 (Level 2)
Suitable for Cohorts: 25/26

Answer ALL EIGHT questions.

Notation: Let $[m] := \{1, \dots, m\}$. We also overload notation so that

$$[\text{pred}] := \begin{cases} 1 & \text{pred is true} \\ 0 & \text{pred is false} \end{cases}$$

Marks for each part of each question are indicated in square brackets.
Calculators are NOT permitted.

1.

- (a) “For polynomial regression, using a higher degree polynomial will always reduce the training error.”
True or False? Explain your answer. [5 marks]
- (b) State the bias-variance decomposition for the expected squared error of a learning algorithm. Briefly explain how increasing model complexity affects each term. [5 marks]

2.

- (a) We consider properties of positive semi-definite matrices.
- (i) Is the following statement true or false? “If K_1 and K_2 are positive semi-definite matrices, then $K_1 + K_2$ is positive semi-definite.” Provide an argument.
- (ii) Is the following statement true or false? “If K is a positive semi-definite matrix, then K^2 (matrix multiplication) is positive semi-definite.” Provide an argument. [5 marks]
- (b) Consider the Gaussian kernel $k(\mathbf{x}, \mathbf{y}) = \exp(-\|\mathbf{x} - \mathbf{y}\|^2)$. Using the fact that the exponential of a kernel is a kernel and that $\|\mathbf{x} - \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\mathbf{x}^\top \mathbf{y}$, argue that the Gaussian kernel is indeed a valid kernel. [10 marks]

3.

- (a) In the SVM dual problem, the constraints include $0 \leq \alpha_i \leq C$. Explain what happens geometrically to a training point $(\mathbf{x}_i, y_i)$ when:

- (i) $\alpha_i = 0$
- (ii) $0 < \alpha_i < C$
- (iii) $\alpha_i = C$

[5 marks]

- (b) The representer theorem states that the solution to kernel ridge regression has the form $f(\mathbf{x}) = \sum_{i=1}^m \alpha_i k(\mathbf{x}_i, \mathbf{x})$. Explain why this is computationally significant when the feature space is infinite-dimensional (e.g., Gaussian kernel). [5 marks]

4.

- (a) In a regression tree, we split a node by choosing variable j and threshold s to minimize:

$$\min_{c_1} \sum_{\mathbf{x}_i \in R_1(j,s)} (y_i - c_1)^2 + \min_{c_2} \sum_{\mathbf{x}_i \in R_2(j,s)} (y_i - c_2)^2$$

What are the optimal values of c_1 and c_2 ? Why is this called a “greedy” algorithm? [5 marks]

- (b) In Random Forests, at each split we only consider a random subset of k features (typically $k = \sqrt{n}$). Explain how this helps compared to standard bagging of decision trees, in terms of the correlation between trees. [5 marks]

5.

- (a) The Weighted Majority algorithm uses parameter $\beta \in [0, 1)$ and updates weights as $w_{t+1,i} = \beta \cdot w_{t,i}$ when expert i makes a mistake.

Explain why setting $\beta = 0$ recovers the Halving algorithm. What is the disadvantage of the Halving algorithm in the non-realizable case? [5 marks]

- (b) For learning k -literal monotone disjunctions, Winnow has mistake bound $O(k \log n)$ while Perceptron has bound $O(kn)$.

Give an intuitive explanation for why Winnow performs better when $k \ll n$ (sparse targets). What property of Winnow’s update rule enables this? [5 marks]

6.

- (a) Write down the graph Laplacian $L = D - W$ for the following weighted graph with 3 vertices and edges: $w_{12} = 2$, $w_{23} = 1$, $w_{13} = 0$. [5 marks]
- (b) For a connected graph, the Laplacian L has eigenvalue 0 with eigenvector $\mathbf{1}$ (all ones). If L has a *second* eigenvector with eigenvalue 0, what can you conclude about the graph structure? Explain your reasoning. [5 marks]

7.

- (a) State the PAC learning framework: what are the inputs to the learner, what guarantee must it provide, and what role do ϵ and δ play? [4 marks]
- (b) For a finite hypothesis class H under the realizability assumption, ERM achieves sample complexity $m \geq \frac{\log(|H|/\delta)}{\epsilon}$. Consider $H =$ all Boolean functions on n binary inputs. What is $|H|$? Using the sample complexity bound, approximately how many samples are needed for $\epsilon = 0.1$, $\delta = 0.1$, and $n = 10$? [6 marks]

8.

- (a) In Online Gradient Descent (OGD), the update rule is $\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla \ell_t(\mathbf{w}_t)$. How is this related to the Perceptron algorithm? Specifically, what loss function and learning rate η recover the Perceptron update? [5 marks]
- (b) The Fixed Share algorithm extends standard expert learning to track a sequence of experts that can switch k times. The regret bound includes a term $k \ln(n - 1)$. Explain intuitively why the regret depends on the number of switches k . If you expect many switches, should you set the share parameter α larger or smaller? [5 marks]

END OF PAPER